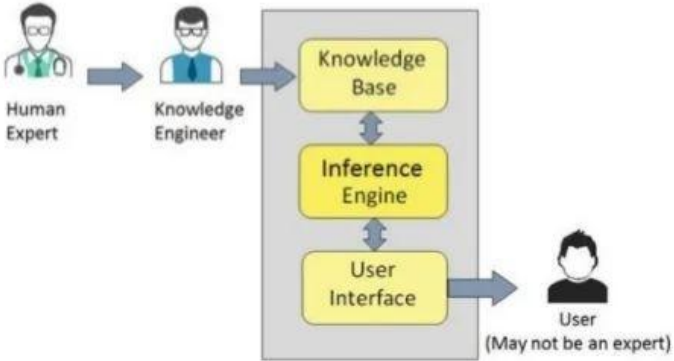
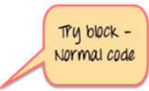


12.1A Artificial intelligence					
Artificial intelligence Artificial intelligence – simulation of human intelligence processes by machines, especially computer systems. Advantages: • Good at detail-oriented jobs. • Reduced time for data-heavy tasks. • AI handles information better than humans. • AI-powered virtual agents are always available. Disadvantages: • Expensive implementation. • Requires deep technical expertise. • Few efficient programmers are available to develop software to implement artificial intelligence. • Only knows what it's been shown.	<i>12.4.3.1 describe spheres where artificial intelligence is applied: industry, education, medicine, gaming industry, society</i> Industry: industrial multitasking robots Education: AI can ensure that educational software is personalized for students. This system emphasizes each student's needs, highlighting specific topics students are weak in and repeating subjects they haven't mastered. AI can help teachers create different content types that make teaching and learning more comfortable. Medicine: AI can help to make better and faster diagnoses. Online virtual <u>health</u> assistants and chatbots to help patients. Gaming industry: AI in gaming is about creating more responsive, adaptive & challenging games. Non-player characters' behavior is determined by artificial intelligence algorithms and engines. The game AI can generate the game landscape or the game world as you go through the game world. Society: Chatbots in banking. Voice assistants. Image/face recognition for face unlock in mobile phones. Social media recommendations.				
Virtual and augmented reality Equipment needed to create the VR experience: • Eye Goggles • Special Gloves • Headphones • Powerful Computer ! AR uses a real-world setting while VR is completely virtual.	<i>12.4.3.2 explain the purpose of virtual and augmented reality</i> Augmented reality (AR) augments your surroundings by adding digital elements to a live view. Virtual reality (VR) is a completely immersive experience that replaces a real-life environment with a simulated one. VR advantages and disadvantages: <table border="1" data-bbox="525 1736 1273 1995"> <thead> <tr> <th data-bbox="525 1736 900 1776">+</th><th data-bbox="900 1736 1273 1776">-</th></tr> </thead> <tbody> <tr> <td data-bbox="525 1776 900 1995"> 1. It is safer, as scenarios can be tried out without endangering life. 2. Potential cost savings as there may be no need to build/use real thing. </td><td data-bbox="900 1776 1273 1995"> 1. Issues with motion sickness. 2. Can cause users to injure themselves if they are not in a large space. </td></tr> </tbody> </table>	+	-	1. It is safer, as scenarios can be tried out without endangering life. 2. Potential cost savings as there may be no need to build/use real thing.	1. Issues with motion sickness. 2. Can cause users to injure themselves if they are not in a large space.
+	-				
1. It is safer, as scenarios can be tried out without endangering life. 2. Potential cost savings as there may be no need to build/use real thing.	1. Issues with motion sickness. 2. Can cause users to injure themselves if they are not in a large space.				

! VR requires a headset device, but AR can be accessed with a smartphone.	3. Doing difficult or impossible tasks;	3. In some cases, users have experienced seizure, so epileptic users must be very cautious.
12.1B Programming paradigms		
Declarative and imperative programming languages Imperative: C, C++, C#, Pascal, Java Declarative: HTML, SCC, SQL, Prolog, XQuery	12.5.1.1 compare declarative and imperative programming languages Imperative languages (procedural languages): These work by typing in lists of instructions that the computer has to follow. Every time the program is run, it follows the set of instructions. Declarative languages: These work by describing what the program should accomplish rather than how it should accomplish it. Imperative – Instruction Declarative – What should be done	
Expert systems An Expert System is an interactive and reliable computer-based decision-making system that uses both facts and heuristics to solve complex decision-making problems. A computer application that solves the most complex issues in a specific domain. Applications of ES: 1. Loan analysis 2. Airline and cargo scheduling 3. Medicine 4. Stock market trading 5. Warehouse optimization ES examples in medicine: MYCIN DENDRAL PXDES CaDet	12.5.1.2 create a simple expert system Characteristics of ES: □ The highest level of expertise. It provides efficiency, accuracy, and imaginative problem-solving. □ Adequate response time. It should be ready to perform within a small amount of time or better than the time taken by an expert. □ Good reliability. The ES must be as reliable as a human expert, it must not make any mistakes. Components of ES:  The diagram illustrates the components of an Expert System. On the left, a 'Human Expert' (represented by a doctor icon) provides input to a 'Knowledge Engineer' (represented by a person icon). The Knowledge Engineer then interacts with a central box containing three components: 'Knowledge Base', 'Inference Engine', and 'User Interface'. The Knowledge Base and Inference Engine are connected by a double-headed arrow, and the Inference Engine and User Interface are also connected by a double-headed arrow. An arrow points from the User Interface to a 'User' (represented by a person icon), who is noted as '(May not be an expert)'. <u>Knowledge Base:</u> stores all the knowledge of an expert. Repository of facts.	

	- Variables are checked to make sure they have been correctly declared and contain the correct data type. - Operations are checked to ensure that they are appropriate for the type of variable being used. <u>Code optimization:</u> - Code optimization makes the program more efficient so it runs faster and uses fewer resources. <u>Code generation:</u> Machine code is generated in this stage. Summary: Lexical Analysis is the first phase when compiler scans the source code. Syntax analysis is all about discovering structure in text. Semantic analysis checks the semantic consistency of the code Once the semantic analysis phase is over the compiler, generate intermediate code for the target machine Code optimization phase removes unnecessary code line and arranges the sequence of statements Code generation phase gets inputs from code optimization phase and produces the page code or object code as a result A symbol table contains a record for each identifier with fields for the attributes of the identifier Error handling routine handles error and reports during many phases https://www.guru99.com/compiler-design-phases-of-compiler.html
12.1C System testing	
Types of test data Testing stage purpose: - To verify that the system meets the requirement of the client - To validate that the system is what the user needs.	<i>12.5.3.1 describe execution errors at programme startup</i> <i>12.5.3.2 perform testing using normal data</i> <i>12.5.3.3 perform testing using extreme data</i> <i>12.5.3.4 perform testing using erroneous data</i> Test data is the data used to run the tests. It has to be accurate and comprehensive so it can expose the defects. Normal data – things that a user is likely to input into the

To improve the quality of product Test plan should include: What is being tested? The test data that will be used. The expected outcome of the test.	program. Extreme (boundary) data - values at the <u>limit</u> of what the program should be able to handle. Erroneous data - inputs that the program <u>should not accept</u>. Alpha testing ✓ Before releasing new software on to the market, developers will test it in-house. Beta testing ✓ The developers will then trial the software with a limited number of users. The users report any problems to the developers. Black box testing is a software testing method in which testers evaluate the functionality of the software without looking at the code structure. White box testing is a method of testing software focusing on the structure of the code and test every possible path through the code. Dry run testing - trace table is used validation - check whether data is reasonable / meets given criteria verification - method to ensure data which is copied / transferred is the same as the original - entering data twice and computer checks both sets of data - check entered data against original document / source
Types of errors <u>Run-time error</u> Sometimes you will have a program that compiles fine but breaks when you actually run it. creating an infinite loop <pre>class connect{ 1 try{ //Code to connect to serv</pre> 	<i>12.5.3.5 describe a syntax error in a programme code</i> <i>12.5.3.6 describe a logic error in a programme code</i> Syntax (compilation) error – an error in the syntax. It may be misspelling words, wrong tokens, missing brackets, semicolon, etc. Logic (semantic) error is when a program compiles, doesn't crash, but the <u>answers</u> that it gives are <u>incorrect</u>. The logic, semantics or meaning, conveyed by the code is wrong. ❑ Exception Handling is a mechanism to handle runtime errors such as ClassNotFoundException, IOException, SQLException, RemoteException, etc.

--	--

PILOTS - 12B

12.2A Information security

Data protection measures

Key threats to data security:

- **Lost/damaged** during a **system crash**
- **corrupted** as a result of faulty disks, power failures
- lost by **accidentally deleting**
- lost or become corrupted by **computer viruses**
- **hacked** into by unauthorized users and deleted/alterd
- destroyed by **natural disasters**

12.1.2.1 describe data and computer systems protection measures such as: physical risks, inter-network screens, information encryption, biometrics, computer virus counter measures

Measures to keep data secure:

- making regular **backups** of files
- running **anti-virus software**
- using strong passwords (8+, lowercase & uppercase letters, numbers and symbols)
- always logging off after use
- using data **encryption** techniques to code data

Encryption

Encryption is the security **method of encoding data** from plaintext to ciphertext, which can only **be decrypted** by the user **with the encryption key**.

Decryption: is the process of converting the cipher text back to plain text.

- Symmetric encryption

uses the same algorithm and key for the encryption and decryption process.



- Asymmetric encryption



Authorization is the **control of a user's access to computer resources**.

Firewall

A firewall can be a hardware device/software program that **controls traffic** between the Internet and a private network/computer system. Firewalls can be customized, and **rules can be set up** that control **which data packets should** be allowed through and which should not be allowed through. Firewall **blocks/filters** traffic that **doesn't meet the criteria/rules**.

A firewall can help **prevent hacking**.

Biometrics means to measure and analyze some human characteristics in order to correctly identify an individual.

12.1.2.2 use means to protect data including back up and disk mirroring

Backup

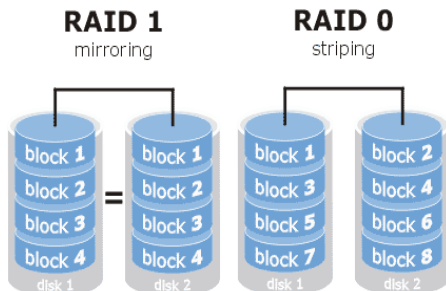
A backup is an additional **copy of data** that can be used to restore and recover data if it was lost/corrupted.

Data mirroring

Redundant Array of Independent Disks (RAID)- is a way of storing the same data in different places on multiple hard disks to protect data in the case of a drive failure.

RAID 0 – striping.
no redundancy, data is stripped

RAID 1 – mirroring.
replication of data to two or more disks



Methods of information systems protection

Copyright – personal non-property and property rights of the author

Copyright – the legal ownership that applies to software, music, films, and other content.

12.1.3.1 analyse ethical problems arising due to computer system cracking

12.1.3.2 analyse problems arising due to malware

Read more about moral, ethical, legal and cultural issues in AQA A-level Computer Science p.300-307

12.1.3.3 explain privacy policies

Two techniques that are used to protect the copyright

Related rights - property rights of a performer, producer of a phonogram, broadcasting, and cable broadcasting organizations, and personal non-property rights of a performer	1. Licensing 2. Digital Rights Management (DRM) Licensing (usually used for software) provides users with paper-based or digital proof that they have purchased software legally and details of what they are allowed to do with the software. Digital Rights Management (DRM) - is the management of legal access to digital content. Examples of DRM: • A movie studio adds software to its DVDs that <u>restricts the making of duplicate copies</u>. • An <u>e-book server</u> confines approach to the <u>copying of</u> material and <u>printing</u> of material • The Law of the Republic of Kazakhstan dated June 10, 1996, No. 6-I On Copyright and Related Rights (as <u>amended and supplemented</u> as of September 23, 2018) Chapter I General Provisions Chapter II Copyright Chapter III Related Rights Chapter IV Collective Management of Property Rights Chapter V Protection of Copyrights and Related Rights
Methods of intellectual property protection EGOV – electronic government Telegram-bot @EgovKzBot	<i>12.1.3.4 analyse problems arising due to disseminating and using information</i> <i>12.1.3.7 use E-gov resources</i> Information service – receiving various information (packages of required documents, phone numbers, addresses of organizations, etc.) 1. Type www.egov.kz in the browser bar and find the information you need Interactive service – receiving various inquiries, notifications, sending statements/appeals 1. Register on the Portal (IIN / BIN, email address) 2. Have an electronic digital signature (EDS) Transaction service – payment of fees, duties in the process of obtaining various certificates, notifications, sending applications/appeals, payment of taxes and penalties for violation of traffic rules To obtain it, you must: 1. Register on the Portal (IIN / BIN, email address)

2. Have a payment card of any bank with the right to **conduct Internet transactions**

SERVICES FOR LIFE SITUATIONS
SERVICES BY STATE AUTHORITIES
COMMERCIAL SERVICES
MAKING A PAYMENT TO THE BUDGET

EDS – ELECTRONIC DIGITAL SIGNATURE

*An **electronic digital signature** is equivalent to the **handwritten signature** of a signatory and entails the **same legal consequences**.*

You can apply for EDS online on the egov portal **www.egov.kz** and on the portal of the National Certification Authority (hereinafter - NEC) **www.pki.gov.kz**.

To successfully submit an application online, you need to install the software: Root certificates of the NTC and Java, which **can be downloaded** at **www.pki.gov.kz**.

After submitting the application online, you will need to appear at the **Public Service Center** (indicated by you when submitting the application) to identify you and provide a completed application (a printed version of the application that you filled out online).

ALGORITHM FOR RECEIVING ANY PAID/FREE ELECTRONIC SERVICE

- If you do not have JAVA installed, download, and install Java distributions at **www.egov.kz/java.zip**
- Launch your internet browser. Enter the address of the e-government portal **www.egov.kz** in the address bar
- Log in using the **IIN (BIN)** and password specified during registration.
- **Select the service** you need and click on the **"Order online service" button**.
- Fill in the **request** fields and **click "Send Request"**.
- Click **Select Certificate**.
- In the window that opens, specify the path to the EDS certificate => My computer => Folder with your certificates => **RSA ...**
- Enter the **password** for the EDS certificate.
- Click **Sign**. Your request has been sent!

	The result of the provision of the service will be automatically saved in the “History of receipt of services” section of the portal (“My Portal”).
12.2B Creating documentation	
Making a text document	<i>12.2.1.2 use headers and footers when documenting a project</i> <i>12.2.1.3 use tables when documenting a project</i> Верхний колонтитул, нижний колонтитул, таблица «вставка» -> «колонтитулы» -> «верхний/нижний» «главная» -> «таблицы» -> «таблица»
Formatting a text document	<i>12.2.1.4 set page parameters when documenting a project</i> <i>12.2.1.5 set page numbers when documenting a project</i> <i>12.2.1.6 set indents and line spacing when documenting a project</i> Нумерация, маркировка «вставка» -> «колонтитулы» -> «номер страницы» «главная» -> «абзац» -> «маркированный список/нумерованный список»
Use resources to check for plagiarism Plagiarism is the act of stealing someone's ideas or works and passing them off as your own.	<i>12.1.3.5 check documents using antiplagiarism resources</i> <i>12.1.3.6 list copyright protection rules</i> Anti-plagiarism is software used to detect/determine if someone has copied someone else’s ideas or works. For example, “etxt”
12.2C System implementation	
Methods of system implementation Implementation stage – the stage of system development in which the actual code, databases, and data structures are created and a system is implemented. Changeover – a change from one system to another (a new one)	<i>12.2.1.7 list system implementation methods</i> <i>12.2.1.8 compare the advantages and disadvantages of system implementation methods</i>

Advantages / Disadvantages of Direct Changeover:

Advantages	Disadvantages
The fastest way of changing from one system to another.	Staff have to be trained and fully ready to use the new system before it is installed. This training can be hard to fit into their daily schedules .
The new system can be used immediately after installation.	All of the old system's data needs to be uploaded into the new system's database before it is installed. Again, this can be time-consuming and difficult to plan .
Cheap method of changeover as you don't pay for two sets of staff. (Only pay the staff who are using the new system)	The old system is completely removed so there is nothing to go back to if the new system fails. Data can be lost permanently .
Less chance of the new system being faulty as it will already have been fully tested and checked.	This can be disastrous and could force the company to shut down for long periods until the system is fixed.

Advantages / Disadvantages of Parallel Running:

Advantages	Disadvantages
If the new system fails, the old system is still available . This way, no data will be lost .	This changeover method is much more expensive . This is because you have the running costs of two systems instead of one (Extra staff to run both systems, electricity etc.)
Staff can be trained in the new system gradually as the changeover doesn't happen immediately. Excellent way to test the new system using live data . (You can compare the results produced by both systems- any difference would mean an error with the new system)	

Advantages / Disadvantages of Pilot Running:

Advantages	Disadvantages
If the new system doesn't work correctly, it only affects one part of the company.	It takes longer to implement the new system across the whole company. (This is because the changeover happens gradually from department to department)
The implementation is on a smaller scale and is easier to manage than direct changeover.	The department taking part in the pilot can lose data if the new system fails. (This is because the new system has replaced the old system)
Staff can be trained gradually from department to department.	
The staff that were involved with the pilot scheme can also help train other staff.	
It is cheaper than parallel running because the pilot is only used in one part of the company.	

Advantages / Disadvantages of Phased Implementation:

Advantages	Disadvantages
Staff can be trained gradually as they only need to train in the part of the system that is currently being phased in .	More expensive than direct changeover as you have to test each phase separately before moving onto the next.
It is easier to find system errors as you are dealing with only one part at a time. (If the whole system is implemented at once, it is harder to pinpoint the error)	Because the system is installed in separate phases, it can take a long time to implement the whole system.
If one part of the new system fails , it is still possible to access the parts of the old system that are still in use .	Only suitable for systems that can be split into separate parts . (Not all systems can be implemented this way)

New system implementation

Implementation plan

- Implementation method
- Reason for method
- Implementation and staff training dates and short explanation
- Suggested training method

12.2.1.9 explain the importance of making a system implementation plan

12.2.1.10 make a system implementation plan

System maintenance

During the use of a system, it may become **necessary to alter the system** for some reason – this is known as **maintenance**.

There are three different types of maintenance

1. **Corrective maintenance**
2. **Adaptive maintenance**
3. **Perfective maintenance**

Corrective	Adaptive	Perfective
Corrective maintenance is necessary when a fault or bug is found	Adaptive maintenance is necessary when conditions change	Perfective maintenance is required to “tweak”

	in the operation of the new system. These are software bugs that were not picked up at the formal testing stage. A technician or the original programmers will be needed to correct the error.	from those that existed when the original system was created. This may be because of a change in the law (tax rates may change, for example) or the hardware may be changed, so that changes need to be made	the system so that it performs better. For example, searching for a particular stock item may be <u>quite slow</u>. The technician decides that supplier details should be stored in another file rather than with the details of the stock, the <u>size of the stock file is reduced</u> and it is far quicker to search. This <u>has not changed how the system operates</u> as far as the user is concerned <u>but the performance improves</u>
--	---	---	---

12.3A Computer systems

OS types

An operating system is a program that controls the execution of application programs and acts as an interface between the user of a computer and the computer hardware.

GUI (Graphical user interface) - a visual interface that allows users to interact with a computer system using graphical elements such as windows, icons, menus and pointers (WIMP).

12.3.1.1 describe features of a real-time operating system

12.3.1.2 describe features of a network operating system

12.3.1.3 describe features of a batch processing operating system

1. **Real-time OS** is one in which processing requests made by the user are executed immediately.
When the computer has to react within a guaranteed time to an input, a real-time operating system (RTOS) is used. A real-time operating system does not necessarily have to be fast. It simply has to be quick enough to respond to inputs in a predictable way. Computers operating in real time are often dedicated to the control of systems such as industrial processes, planes and space flights.
Example: Airline traffic control systems, Command Control Systems, Airlines reservation system
2. **Network OS** – software that connects multiple devices and computers on the network and allows them to share resources on the network.
Example: Microsoft Windows Server 2003, UNIX, Linux, Mac OS X
3. **Batch processing OS** is a computer OS that processes large amounts of data in batches. This type of system is typically used by businesses and organizations that need to process large amounts of data quickly and efficiently.
Example: Payroll System, Bank Statements.
This method of operation lends itself to jobs with similar inputs, processing, and outputs where no human intervention is needed. Jobs are stored in a queue until the computer is ready to deal with them. Often batch processed jobs are done overnight.

12.3.1.4 *define the advantages and disadvantages of a graphical user interface (GUI)*

ADVANTAGES	DISADVANTAGES
Easy to use, especially for a beginner	GUIs take up a much larger amount of hard disk space than other interfaces
You do not have to learn complicated commands	They can be slow for experienced programmers to use.
They enable you to easily exchange information between software using cut and paste or 'drag and drop'.	GUIs may not be portable between different operating systems, which can make it challenging to use the same interface on different devices.
GUIs allow users to perform multiple tasks simultaneously,	

such as running multiple applications, moving files, and accessing the internet.	
GUIs are designed to be accessible to people with disabilities, including those who are visually impaired. Special features such as screen readers and magnifiers help them to interact with the system.	

- When discussing user interfaces, it is important to note that **Windows XP, Windows Vista, Apple OSX, and Ubuntu** all have graphical user interfaces.

12.3.1.5 define the advantages and disadvantages of a command line interface (CLI)

CLI (Command line interface) is a text-based interface that allows users to interact with a computer system by typing commands into a terminal or shell.

ADVANTAGES	DISADVANTAGES
If the user knows the correct commands then this type of interface can be much faster than any other type of interface	For someone who has never used a CLI, it can be very confusing
Needs much less memory and processing power	Commands have to be typed precisely. If there is a spelling error the command will fail
A low resolution, cheaper monitor can be used with this type of interface	There are a large number of commands which need to be learned
Does not use as much CPU processing time as others	
Does not require Windows to run	

Natural-language user interface (LUI or NLUI) is a type of computer-human interface where linguistic phenomena are used for creating, selecting, and modifying data in software applications.

Examples: Alisa, Siri, Alexa, Google Assistant or Cortana

Gesture-based user interface refers to using specific physical gestures in order to operate an interface.

Gesture recognition uses computer sensors to detect and understand human gestures and movements. Gestures let users interact with screen elements using touch.

An example of a common command-driven interface is **MS-DOS**. The MS-DOS command to display all files on c:\ would be: dir c:\

1. Other examples: Terminal on macOS and Linux
2. Command Prompt on Windows
3. PowerShell on Windows
4. MySQL Command Line Client
5. Git Bash

Commands on Windows:

Step	CLI Command	Description
1	Drive:	Switching Drives(eg. D:)
2	Mkdir	Making a folder(eg. mkdir test)
3	cd	Moving into a directory (eg. Cd test)
4	Dir	List files in the <u>current directory</u>
5	notepad testing.bat	Create a batch file name:testing.bat (eg., notepad testing.bat)
6		Write the following in the testing.bat file Mspaint.exe Calc.exe Save the file as testing.bat file in the d:\test
7	Testing.bat	Executing batch file

12.3.1.6 define the advantages and disadvantages of natural-language and gesture-recognition user interfaces

NL UI:

+	-
+ The user does <u>not need to be trained</u> on how to use the interface	- The interface can <u>only respond to programmed commands</u>
+ Suitable for people with <u>disabilities</u>	- Very <u>difficult in programming</u> , so it is used when other types of interface are not suitable
+ Allows interaction at a <u>distance</u>	- The voice interface may need training as <u>different users</u> have <u>different pronunciations</u> .

GR UI:

ADVANTAGES	DISADVANTAGES
+ It is an intuitive and <u>natural way of interaction</u> .	- Irrelevant objects with hands can mislead the recognition system.
+ Recognizes both <u>static and dynamic</u> hand movements. + It has minimal hardware requirements.	- The performance of this system drops as the distance between user and camera increases.
+ Easily be implemented in real-time systems.	- It still doesn't produce an interface that can replace physical controllers.

	<table border="1"> <tr> <th data-bbox="448 129 852 210">Natural-language user interface can be used in:</th><th data-bbox="852 129 1272 210">Gesture-based user interface: types</th></tr> <tr> <td data-bbox="448 210 852 981"> Search engines User can input sentences in a natural language and get precise search results. Computer Games User can interact with game characters as naturally as with intellectual human beings enhancing his game play experience. Document management software Natural Language Tools ensure high-quality information analysis by employing semantic document comparison, document summarization, document categorization, and sentiment analysis. </td><td data-bbox="852 210 1272 981"> When it comes to classifying different types of gestures, Material, Design defines three types: Navigational gestures Gestures that help users move through a product easily. Action gestures Action gestures can perform actions such as scrolling. Transform gestures Gestures that allow users to transform objects such as an element's size, position, and rotation using gestures. </td></tr> </table>	Natural-language user interface can be used in:	Gesture-based user interface: types	Search engines User can input sentences in a natural language and get precise search results. Computer Games User can interact with game characters as naturally as with intellectual human beings enhancing his game play experience. Document management software Natural Language Tools ensure high-quality information analysis by employing semantic document comparison, document summarization, document categorization, and sentiment analysis.	When it comes to classifying different types of gestures, Material, Design defines three types: Navigational gestures Gestures that help users move through a product easily. Action gestures Action gestures can perform actions such as scrolling. Transform gestures Gestures that allow users to transform objects such as an element's size, position, and rotation using gestures.												
Natural-language user interface can be used in:	Gesture-based user interface: types																
Search engines User can input sentences in a natural language and get precise search results. Computer Games User can interact with game characters as naturally as with intellectual human beings enhancing his game play experience. Document management software Natural Language Tools ensure high-quality information analysis by employing semantic document comparison, document summarization, document categorization, and sentiment analysis.	When it comes to classifying different types of gestures, Material, Design defines three types: Navigational gestures Gestures that help users move through a product easily. Action gestures Action gestures can perform actions such as scrolling. Transform gestures Gestures that allow users to transform objects such as an element's size, position, and rotation using gestures.																
CPU architectures	<i>12.3.2.1 describe the RISC architecture</i> <i>12.3.2.2 describe the CISC architecture</i> <i>12.3.2.3 compare RISC and CISC</i> <table border="1"> <thead> <tr> <th data-bbox="432 1171 959 1238">CISC</th><th data-bbox="959 1171 1485 1238">RISC</th></tr> </thead> <tbody> <tr> <td data-bbox="432 1238 959 1312">Complex Instruction Set</td><td data-bbox="959 1238 1485 1312">Reduced Instruction Set.</td></tr> <tr> <td data-bbox="432 1312 959 1417">Non-fixed size of instruction, e.g. instructions have variable format.</td><td data-bbox="959 1312 1485 1417">Fixed-size of instruction</td></tr> <tr> <td colspan="2" data-bbox="432 1417 959 1765"> <table> <tr> <th>CISC</th><th>RISC</th></tr> <tr> <td> <pre>***** ***** ***** ***** ***** ***** *****</pre> </td><td> <pre>***** ***** ***** ***** ***** ***** *****</pre> </td></tr> <tr> <td>non-fixed size of instruction</td><td>fixed size of instruction</td></tr> </table> </td></tr> <tr> <td data-bbox="432 1765 959 1944">Uses complex instructions each of which may take multiple cycles.</td><td data-bbox="959 1765 1485 1944">Complex tasks can only be performed by combining several simple tasks so a task may take several machine cycles.</td></tr> </tbody> </table>	CISC	RISC	Complex Instruction Set	Reduced Instruction Set.	Non-fixed size of instruction, e.g. instructions have variable format.	Fixed-size of instruction	<table> <tr> <th>CISC</th><th>RISC</th></tr> <tr> <td> <pre>***** ***** ***** ***** ***** ***** *****</pre> </td><td> <pre>***** ***** ***** ***** ***** ***** *****</pre> </td></tr> <tr> <td>non-fixed size of instruction</td><td>fixed size of instruction</td></tr> </table>		CISC	RISC	<pre>***** ***** ***** ***** ***** ***** *****</pre>	<pre>***** ***** ***** ***** ***** ***** *****</pre>	non-fixed size of instruction	fixed size of instruction	Uses complex instructions each of which may take multiple cycles.	Complex tasks can only be performed by combining several simple tasks so a task may take several machine cycles.
CISC	RISC																
Complex Instruction Set	Reduced Instruction Set.																
Non-fixed size of instruction, e.g. instructions have variable format.	Fixed-size of instruction																
<table> <tr> <th>CISC</th><th>RISC</th></tr> <tr> <td> <pre>***** ***** ***** ***** ***** ***** *****</pre> </td><td> <pre>***** ***** ***** ***** ***** ***** *****</pre> </td></tr> <tr> <td>non-fixed size of instruction</td><td>fixed size of instruction</td></tr> </table>		CISC	RISC	<pre>***** ***** ***** ***** ***** ***** *****</pre>	<pre>***** ***** ***** ***** ***** ***** *****</pre>	non-fixed size of instruction	fixed size of instruction										
CISC	RISC																
<pre>***** ***** ***** ***** ***** ***** *****</pre>	<pre>***** ***** ***** ***** ***** ***** *****</pre>																
non-fixed size of instruction	fixed size of instruction																
Uses complex instructions each of which may take multiple cycles.	Complex tasks can only be performed by combining several simple tasks so a task may take several machine cycles.																

	Many instructions and <u>addressing modes</u> are available. <u>Can perform complex tasks faster.</u>	A <u>limited number of instructions</u> are available. The instruction performs a simple task. <u>Can perform simple tasks faster.</u>
	<u>Small code sizes</u>	Large code sizes
	High cycles per second	Low cycles per second
	Has a <u>complicated</u> and <u>expensive processor design.</u>	<u>Simple processor</u> design.
	Has <u>more components</u> on the CPU than a RISC processor	<u>Less components</u> on CPU than in CISC processor
	Use <u>less RAM</u>	Use more RAM
	Produces more heat, requires heat sink/fan	Produces less heat
	Used in PC	Used in tablets and mobile devices
	Memory-to-memory	Register-to-register
	Require <u>more power</u>	Require <u>less power</u>
	Uses <u>less registers</u> than in RISC	Uses <u>more registers</u> than in CISC
	Cannot use pipelining	Uses <u>pipelining</u>
Memory addressing principle Addressing Modes —the way in which the operand of an instruction is specified. The addressing mode specifies a rule for interpreting or modifying the address field of the instruction before the operand is actually executed.	<i>12.3.4.1 explain memory addressing</i> Memory addressing is the process of identifying the location of a particular piece of data in computer memory. Direct addressing – the contents of the memory location in the operand are used. For example, if the memory location with address 200 contained the value 20, the assembly language instruction LDD 200 would store 20 in the accumulator. Absolute and direct addressing are the same. Indirect addressing – the contents of the contents of the memory location in the operand are used. For example, if the memory location with address 200 contained the value 20 and the memory location with address 20 contained the value 5, the assembly language instruction LDI 200 would store 5 in the accumulator. Indexed addressing – the contents of the memory location found by adding the contents of the index register (IR) to the address of the memory location in the operand are used.	

For example, if IR contained the value 4 and memory location with address 204 contained the value 17, the assembly language instruction LDX 200 would store 17 in the accumulator.

Immediate addressing – the value of the operand only is used.

For example, the assembly language instruction LDM #200 would store 200 in the accumulator.

Relative addressing – the memory address used is the current memory address added to the operand.

For example, JMR #5 would transfer control to the instruction 5 locations after the current instruction.

12.3.4.2 explain principle of memory organization by segment and page

Segmentation – a memory management technique that breaks up physical memory into fixed-size blocks called "pages."

Paging – form of memory management which divides up physical memory and logical memory into fixed-size memory blocks.

Paging	Segmentation
a page is a fixed-size block of memory	a segment is a variable-size block of memory
since the block size is fixed, it is possible that all blocks may not be fully used – this can lead to internal fragmentation	because memory blocks are a variable size, this reduces risk of internal fragmentation but increases the risk of external fragmentation
the user provides a single value – this means that the hardware decides the actual page size	the user will supply the address in two values (the segment number and the segment size)
a page table maps logical addresses to physical addresses (this contains the base address of each page stored in frames in physical memory space)	segmentation uses a segment map table containing segment number + offset (segment size); it maps logical addresses to physical addresses
the process of paging is essentially invisible to the user/programmer	segmentation is essentially a visible process to a user/programmer
procedures and any associated data cannot be separated when using paging	procedures and any associated data can be separated when using segmentation
paging consists of static linking and dynamic loading	segmentation consists of dynamic linking and dynamic loading

Similarities		Differences	
1	Both allow programs to run despite insufficient memory.	Pages are fixed sized. Segments are different sizes.	1
2	Pages and segments are stored on disk.	Pages are made to fit sections of memory. Segments are complete sections of programs.	2
3	Pages and segments are transferred into memory when needed.	Pages are physical divisions. Segments are logical divisions.	3

12.3.4.3 explain the virtual machine concept

Virtual machine - an emulation of an existing computer system. A computer

Emulation – the use of an app/device to imitate the behaviour of another program/device; for example, running an OS on a computer which is not normally compatible.

Host OS – an OS that controls the physical hardware.

Guest OS – an OS running on a virtual machine.

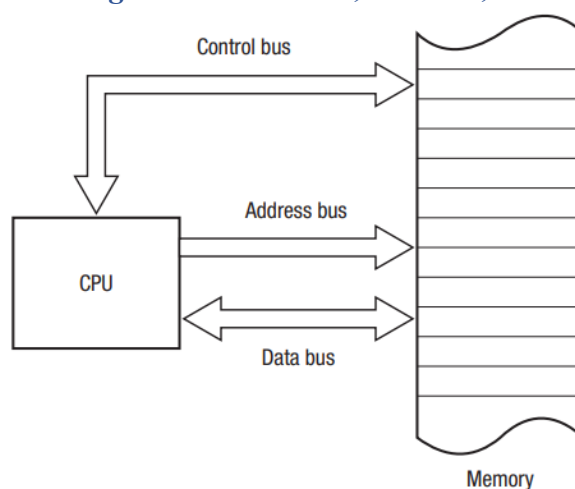
Hypervisor – virtual machine software that creates and runs virtual machines. Suppose you are using a PC running in a Windows 10 environment and you have downloaded some software that will only run on an Apple computer. It is possible to emulate the running of the Apple software on the Windows PC (hardware) – this is known as a virtual machine. The emulation will allow the Apple software to use all of the hardware on the host PC.

Effectively, a virtual machine runs the existing OS (called the host operating system) and oversees the virtual hardware using a guest operating system – in our example above, the host OS is Windows 10 and the guest OS is Apple. The emulation engine is referred to as a hypervisor; this handles the virtual hardware (CPU, memory, HDD and other devices) and maps them to the physical hardware on the host computer.

+	-
• Less electricity used - you use one computer instead of two • You can quickly switch between operating systems • Cheaper, you don't need to buy a new computer • You can store the entire "computer" in one file	• Makes your computer slower by starting two systems at the same time • If the main operating system crashes, you will lose other operating systems as well • There may be compatibility issues

System bus

12.3.2.4 explain how data are transferred between the components of a computer system through the address bus, data bus, and control bus



Address bus:

- Used to carry the address to memory and IO.
- Unidirectional.

	- Based on width of a address bus we can determine the capacity of a main memory Data bus: - Used to carry the binary data between the CPU, memory and IO. - Bidirectional. - Based on the width of a data bus we can determine the word length of a CPU. - Based on the word length we can determine the performance of a CPU. Control bus: - Used to carry the control signals and timing signals - Bidirectional - Control signals indicates type of operation. - Timing Signals used to synchronize the memory and IO operations with a CPU clock.
Fetch-execute cycle Fetch-execute cycle -the continuous process carried out by the processor when running programs	<i>12.3.2.5 explain the instruction cycle (fetch/decode/execute)</i> Fetch – the processor fetches the program’s next instruction from memory. The instruction will be stored at a memory address and will contain the instruction in binary code. Decode – the processor works out what the binary code at that address means. Execute – the processor carries out the instruction which may involve reading an item f data from memory, performing a calculation or writing data back into memory. Program Counter (PC) - an incrementing counter that keeps track of the memory address of which instruction is to be executed next. Memory Address Register (MAR) - the address in main memory that is currently being read or written. Memory Buffer Register (MBR) or Memory Data Register (MDR) - a two-way register that holds data fetched from memory (and ready for the CPU to process) or data waiting to be stored in memory. Current Instruction register (CIR) - a temporary holding ground for the instruction that has just been fetched from memory. Status Register (SR) - contains bits that are set or cleared depending on the result of an instruction. For example, one bit will be set if an overflow has occurred, other bits will indicate whether the result of the last instruction was negative, zero or caused a carry. Accumulator (ACC) - an internal CPU register used as the default location to store any calculations performed by the arithmetic and logic unit. <i>12.3.2.6 explain how the clock rate, word length and bus width affect the CPU performance</i> Word length: word size refers to the amount of data that can be handled at one time by the processor. The larger the word size, the greater the amount of data that can be transferred to the CPU in one pass. This is an important factor for the processor performance, as to process an instruction it might well require multiple fetches of data from the main memory. Therefore, being able to pass larger amounts of data to the processor with every pass, the system is likely to carry out instructions faster.

Bus width: The **width of the data bus** determines the number of bits that can be transferred to or from in one operation (i.e. at the same time, in one pass). The larger the data bus, the better the processor performance. This is because the greater the width of the data bus, the more data can be transferred between the internal components simultaneously.

Clock speed: The clock speed of a CPU is the rate at which it can execute instructions, measured in GHz (gigahertz). A higher clock speed generally means that the CPU can execute instructions more quickly, resulting in better performance.

Cache size: The CPU cache is a small amount of memory that is located on the CPU itself, and is used to store frequently accessed data and instructions. A larger cache can improve CPU performance by reducing the amount of time it takes to access frequently used data.

Number of cores: Modern CPUs often have multiple cores, which allow them to execute multiple instructions simultaneously. CPUs with more cores can generally perform better for tasks that can be parallelized, such as rendering graphics or running multiple applications at once.

Instruction set architecture: The instruction set architecture (ISA) of a CPU determines the types of instructions it can execute. A more advanced ISA can enable a CPU to perform more complex operations, leading to better performance.

Boolean logic

(you can find more info on this topic in the document called “11 grade, term 1”)

12.3.3.1 reduce formulas to normal logic using the laws of formal logic and rules of logical transformation

12.3.3.2 simplify logical expressions using the laws of logic

12.3.3.3 build logical structures

12.3.3.4 build logical expressions according to logical structures

Name	AND form	OR form
Identity law	$1A = A$	$0 + A = A$
Null law	$0A = 0$	$1 + A = 1$
Idempotent law	$AA = A$	$A + A = A$
Inverse law	$A\bar{A} = 0$	$A + \bar{A} = 1$
Commutative law	$AB = BA$	$A + B = B + A$
Associative law	$(AB)C = A(BC)$	$(A + B) + C = A + (B + C)$
Distributive law	$A + BC = (A + B)(A + C)$	$A(B + C) = AB + AC$
Absorption law	$A(A + B) = A$	$A + AB = A$
De Morgan's law	$\overline{AB} = \bar{A} + \bar{B}$	$\overline{A + B} = \bar{A}\bar{B}$

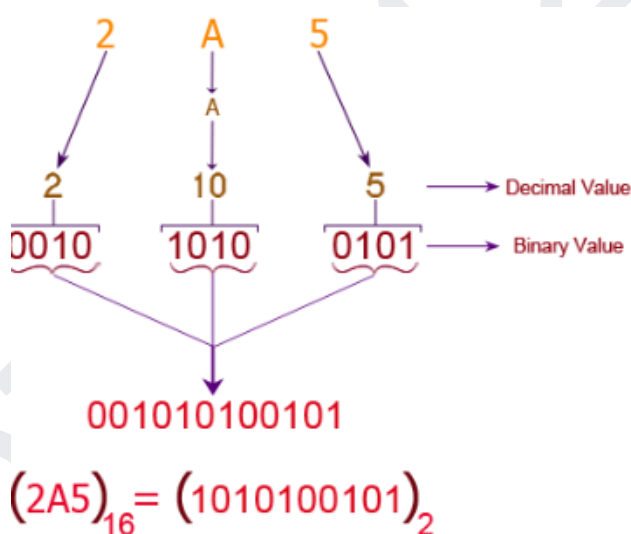
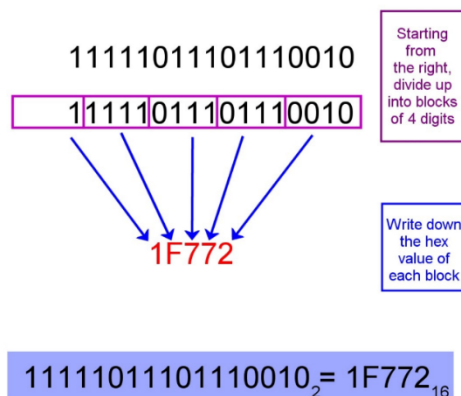
12.3B Computer systems

Number representation methods

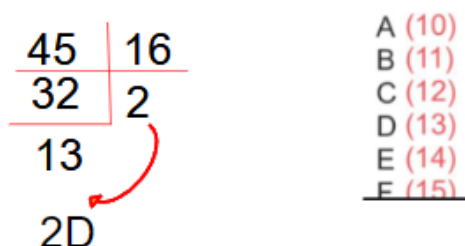
Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

12.1.1.1 convert binary numbers to hexadecimal

Converting Binary to Hex



Denary to Hexadecimal



12.1.1.2 explain the advantages of using hexadecimal numbers in computer systems

- 1. Compact representation:** Hexadecimal numbers can represent a large range of values with a small number of digits. For example, one byte (8 bits) can be represented by two hexadecimal digits. This compact representation makes it easier to read and write numbers in a computer system.
- 2. Easy conversion:** Converting between hexadecimal and binary is easy

	because each hexadecimal digit corresponds to a group of four binary digits. This makes it <u>easy to work with binary data in a human-readable format</u>. 3. Color representation: Hexadecimal numbers are commonly used to <u>represent colors</u> in computer graphics. Each color component (red, green, and blue) can be represented by a two-digit hexadecimal number, making it easy to specify colors in code. 4. Error detection: Hexadecimal numbers can be useful for error detection in data transmission. This is because a single-digit error in a hexadecimal number corresponds to a single-bit error in the binary data it represents, which can be easily detected.																																															
Addition and multiplication of binary numbers	<i>12.1.1.3 perform the arithmetic operations: addition and multiplication of binary numbers</i> Addition: <table><tr><td><pre>01011 (11₁₀) +00001 (1₁₀) ----- 01100 (12₁₀) 11</pre></td><td><table><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td></td></tr><tr><td>+0</td><td>+0</td><td>+1</td><td>+1</td><td>+1</td><td>+1</td><td></td></tr><tr><td>00</td><td>01</td><td>01</td><td>10</td><td>11</td><td>100</td><td></td></tr><tr><td></td><td></td><td></td><td>↑</td><td>↑</td><td>↑</td><td></td></tr><tr><td></td><td></td><td></td><td colspan="3">carried bit</td><td></td></tr></table></td></tr></table> Multiplication: <table><tr><td>101</td><td></td></tr><tr><td>x 10</td><td></td></tr><tr><td>----</td><td></td></tr><tr><td>0</td><td>= 0 * 101</td></tr><tr><td>1010</td><td>= 10 * 101 [or in denary 2 * 5 = 10]</td></tr></table> <i>12.1.1.4 represent positive and negative numbers in binary using complement in n-bit range</i>	<pre>01011 (11₁₀) +00001 (1₁₀) ----- 01100 (12₁₀) 11</pre>	<table><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td></td></tr><tr><td>+0</td><td>+0</td><td>+1</td><td>+1</td><td>+1</td><td>+1</td><td></td></tr><tr><td>00</td><td>01</td><td>01</td><td>10</td><td>11</td><td>100</td><td></td></tr><tr><td></td><td></td><td></td><td>↑</td><td>↑</td><td>↑</td><td></td></tr><tr><td></td><td></td><td></td><td colspan="3">carried bit</td><td></td></tr></table>	0	1	0	1	1	1		+0	+0	+1	+1	+1	+1		00	01	01	10	11	100					↑	↑	↑					carried bit				101		x 10		----		0	= 0 * 101	1010	= 10 * 101 [or in denary 2 * 5 = 10]
<pre>01011 (11₁₀) +00001 (1₁₀) ----- 01100 (12₁₀) 11</pre>	<table><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td></td></tr><tr><td>+0</td><td>+0</td><td>+1</td><td>+1</td><td>+1</td><td>+1</td><td></td></tr><tr><td>00</td><td>01</td><td>01</td><td>10</td><td>11</td><td>100</td><td></td></tr><tr><td></td><td></td><td></td><td>↑</td><td>↑</td><td>↑</td><td></td></tr><tr><td></td><td></td><td></td><td colspan="3">carried bit</td><td></td></tr></table>	0	1	0	1	1	1		+0	+0	+1	+1	+1	+1		00	01	01	10	11	100					↑	↑	↑					carried bit															
0	1	0	1	1	1																																											
+0	+0	+1	+1	+1	+1																																											
00	01	01	10	11	100																																											
			↑	↑	↑																																											
			carried bit																																													
101																																																
x 10																																																

0	= 0 * 101																																															
1010	= 10 * 101 [or in denary 2 * 5 = 10]																																															

Signed binary numbers

If the MSB is 0 then the number is positive, if 1 then the number is negative.

0000 0101 (positive)
1111 1011 (negative)

Representing NEGATIVE number using Two's complement:

Steps:

- Convert denary to binary (usual method)
- Add 0 to the front (left side)
- Flip every bit
- Add 1 to the last bit

Example: -25

- 11001
- 011001
- 100110
- 100111

Representing POSITIVE number using Two's complement:

Steps:

- Convert denary to binary (usual method)
- Add 0 to the front (left side)

Example: +25

- 11001
- 011001

12.1.1.5 perform complement subtraction

X (82 denary) 0101 0010
Y (78 denary) 0100 1110 -

An easier way to subtract Y from X is to **add** the negative value of Y to the value of X

$$X - Y = X + (-Y)$$

To do this we first need to find the negative value of Y (78 denary)

0100 1110 find the right most one

0100 1110

1011 0010 flip all the bits to its left

Now try the sum again

0101 0010 X(82 denary)

1011 0010 + Y(-78 denary)

0000 0100

(¹)¹¹¹ 1 the one carried over the bit 9 is ignored

Which comes out as:

128	64	32	16	8	4	2	1
0	0	0	0	0	1	0	0
					4		

= 4 = 82-78

12.1.1.6 use binary numbers with a fixed point to represent fractional numbers with a given number of bits

12.1.1.7 represent positive and negative floating-point decimal numbers in binary

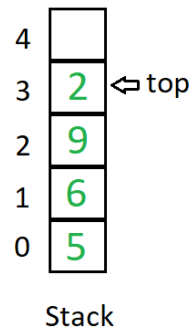
Binary fraction	Fraction	Decimal fraction
0.1	1/2	0.5
0.01	1/4	0.25
0.001	1/8	0.125
0.0001	1/16	0.0625
0.00001	1/32	0.03125
0.000001	1/64	0.015625

Stack and queues

12.5.2.1 describe the operation of stack and queue data structures

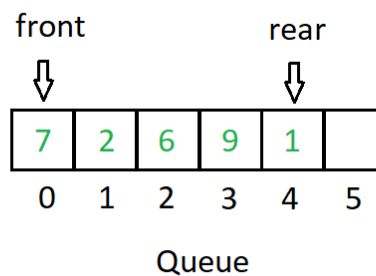
A **stack** is a linear data structure that follows the "last-in, first-out" (**LIFO**) principle. This means that the last element added to the stack is the first one to be removed.

The insertion of an element into the stack is called **push** operation, and the deletion of an element from the stack is called **pop** operation. In stack, we always keep track of the last element present in the list with a pointer called **top**.



A **queue** is a linear data structure that follows the "first-in, first-out" (**FIFO**) principle. This means that the first element added to the queue is the first one to be removed.

The insertion of an element in a queue is called an **enqueue** operation and the deletion of an element is called a **dequeue** operation.

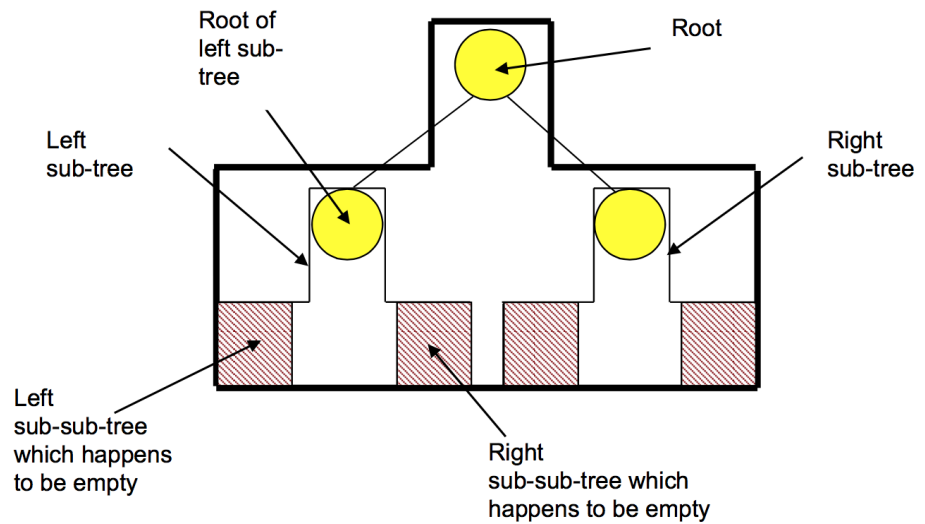


Binary tree

12.5.2.2 build a binary tree

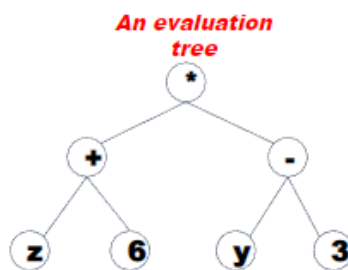
A **binary tree** is an abstract data type defining a finite set of elements.

Figure 4 Division of a binary tree into  a root, left sub-tree and right sub-tree



WHY BINARY TREES ARE USEFUL?

- They can represent the order in which mathematical expressions are evaluated
 - **For example:** $(z + 6) * (y - 3)$
- The expression in the tree is evaluated from the bottom up and from left to right.



Evaluation order

$z + 6$	intermediate result 1
$y + 3$	intermediate result 2
intermediate result 1 *	
intermediate result 2	final result

	Examples: • pre-order DAE • in-order ADE • post-order AED • pre-order DBACFEG • in-order ABCDEFG • post-order ACBEGFD • pre-order +*AC/EG • in-order A*C+E/G (order of evaluation is (A*C)+(E/G)) • post-order AC*EG/+ (Reverse Polish form)
12.3C Communication and networks	
Internet and the World Wide Web	<i>12.6.2.1 distinguish features of the World Wide Web, Intranet and the Internet</i> World Wide Web (WWW): The WWW is a <u>collection of interconnected documents and web pages</u> that are <u>accessed through the Internet</u>. The WWW uses the Hypertext Transfer Protocol (HTTP) to transfer data between servers and clients, and it is based on a client-server model. The main features of the WWW include: • Hyperlinks: The ability to link pages and documents together using hyperlinks is a key feature of the WWW. • Web browsers: Web browsers, such as Google Chrome and Mozilla Firefox, are used to access and navigate the web. • Uniform Resource Locator (URL): URLs are used to specify the location of a resource on the web. • Multimedia: The WWW supports the use of multimedia elements such as images, audio, and video.

Intranet: An Intranet is a private network that is used within an organization to share information and resources among employees. An Intranet can be accessed through a web browser, but it is typically not connected to the Internet.

The main features of an Intranet include:

- Private access: An Intranet is not accessible to the general public and requires a login and password to access.
- Collaboration tools: An Intranet often includes tools for collaboration, such as shared calendars, document sharing, and chat.
- Security: An Intranet is typically protected by firewalls and other security measures to prevent unauthorized access.

Internet: The Internet is a global network of interconnected computer networks that allows for the transfer of data and information between devices. The Internet is the underlying infrastructure that supports the WWW and many other applications.

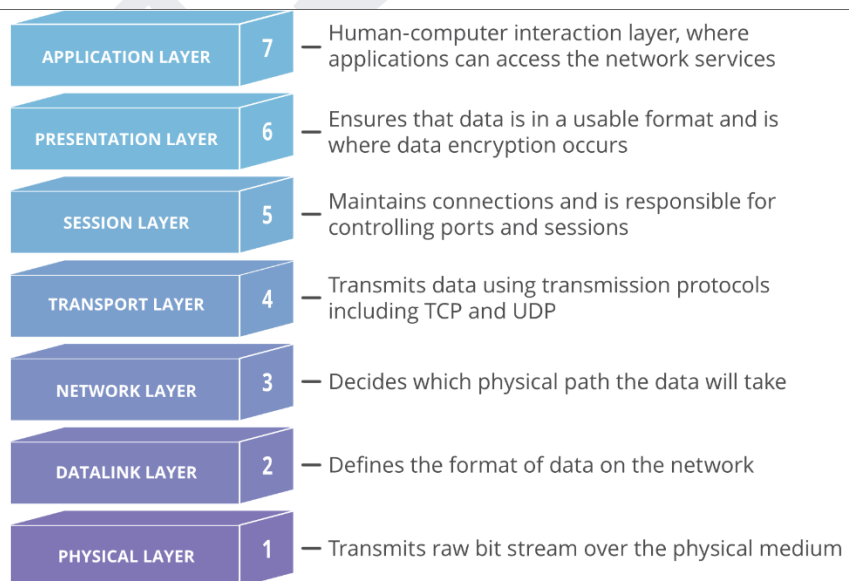
The main features of the Internet include:

- Wide accessibility: The Internet is accessible from almost anywhere in the world.
- Public access: The Internet is a public network and anyone can access it without a login or password.
- Various applications: The Internet supports many different applications, such as email, instant messaging, file sharing, and e-commerce.
- Security risks: The Internet is vulnerable to security risks, such as hacking, phishing, and malware.

OSI models

The OSI (Open Systems Interconnection) network model is a theoretical framework that outlines the functions of a communication system.

12.6.1.2 describe the functions of the OSI network model layers



7) Application Layer: This layer provides an interface between the user and the network. It supports applications such as email, web browsing, file transfer, and remote access. It also provides services such as network virtual terminal, file transfer access and management, and network printing.

Keywords: user interface, application protocols, data exchange, file transfer

Common protocols: HTTP, FTP, SMTP, POP3, IMAP, Telnet, SSH

6) Presentation Layer: This layer is responsible for data representation and encryption. It transforms data into a format that can be understood by the application layer and provides encryption and decryption services.

Keywords: data formats, data conversion, encryption, decryption

Common protocols: SSL, TLS, MIME

5) Session Layer: This layer establishes, maintains, and terminates connections between applications running on different devices. It also provides services such as checkpointing and recovery to ensure reliable communication.

Keywords: sessions, synchronization, checkpoints, recovery

Common protocols: NetBIOS, PPTP, RPC

4) Transport Layer: This layer provides reliable end-to-end communication between devices by establishing a connection-oriented or connectionless communication. It also provides flow control, error recovery, and data segmentation.

Keywords: connections, segments, flow control, error recovery, reliability

Common protocols: TCP, UDP, SCTP

3) Network Layer: This layer is responsible for the logical addressing and routing of data packets between devices on different networks. It determines the best path for data transmission and handles congestion control.

Keywords: logical addressing, routing, packets, IP addresses, congestion control

Common protocols: IP, ICMP, ARP, RARP, OSPF, BGP

2) Data Link Layer: This layer provides reliable transmission of data over the physical layer by detecting and correcting errors, managing flow control, and organizing data into frames. It also handles the addressing of devices on the network.

Keywords: frames, data link addressing, error detection and correction, flow control

Common protocols: Ethernet, Wi-Fi, Token Ring, HDLC, PPP

1) Physical Layer: This layer deals with the physical connections between devices, such as the cables, connectors, and signals that transmit data across the network. Its functions include signal transmission, data encoding and decoding, and physical addressing.

Keywords: transmission media, bits, signals, physical addressing, encoding, decoding

	Common protocols: Ethernet, Wi-Fi, Bluetooth, USB, RS-232														
Packet switching and circuit switching	<i>12.6.1.1 explain the difference between packet switching and circuit switching</i> Packet switching is a communication method where data is divided into smaller units called packets and transmitted over the network. Each packet contains the source and destination addresses, as well as other information needed for routing. The packets may take different paths to reach their destination, and they may be transmitted <u>out of order</u> or delayed due to network congestion. Circuit switching is a communication method where a dedicated communication path, or circuit, is established between two devices before data transmission begins. The circuit remains dedicated to the communication for the duration of the session, and no other devices can use it while the session is in progress. Circuit switching is commonly used in voice communication and some types of data communication. <table border="1"> <thead> <tr> <th><u>Circuit Switching</u></th><th><u>Packet Switching</u></th></tr> </thead> <tbody> <tr> <td>Circuit switching requires a dedicated path before sending data from source to destination.</td><td>Packet switching does not require any dedicated path to send data from source to destination.</td></tr> <tr> <td>It reserves the entire bandwidth in advance.</td><td>It does not reserve bandwidth in advance</td></tr> <tr> <td>No store and forward transmission</td><td>It supports store and forward transmission</td></tr> <tr> <td>Each packet follows the same route</td><td>A packet can follow any route</td></tr> <tr> <td>Call setup is required</td><td>No call setup is require</td></tr> <tr> <td>Bandwidth wastage</td><td>No bandwidth wastage</td></tr> </tbody> </table>	<u>Circuit Switching</u>	<u>Packet Switching</u>	Circuit switching requires a dedicated path before sending data from source to destination.	Packet switching does not require any dedicated path to send data from source to destination.	It reserves the entire bandwidth in advance.	It does not reserve bandwidth in advance	No store and forward transmission	It supports store and forward transmission	Each packet follows the same route	A packet can follow any route	Call setup is required	No call setup is require	Bandwidth wastage	No bandwidth wastage
<u>Circuit Switching</u>	<u>Packet Switching</u>														
Circuit switching requires a dedicated path before sending data from source to destination.	Packet switching does not require any dedicated path to send data from source to destination.														
It reserves the entire bandwidth in advance.	It does not reserve bandwidth in advance														
No store and forward transmission	It supports store and forward transmission														
Each packet follows the same route	A packet can follow any route														
Call setup is required	No call setup is require														
Bandwidth wastage	No bandwidth wastage														
Packet routing A network card, network adapter or NIC (network interface controller) is an electronic hardware component designed to allow computers to communicate over a computer network.	<i>12.6.2.2 describe the role of MAC addresses in packet routing</i> MAC address (Media Access Control address, also known as the <u>Physical Address</u> of a network device) - a unique identifier assigned to a network interface controller (NIC) for use as a network address in communications within a network segment. This use is common in most IEEE 802 networking technologies, including Ethernet, Wi-Fi, and Bluetooth. The format of MAC address: MAC Address is a 12-digit <u>hexadecimal</u> number (6-Byte binary number), which is mostly represented by Colon-Hexadecimal notation. The First 6 digits (say 00:40:96) of the MAC Address identify the manufacturer, called OUI (Organizational Unique Identifier). The remaining digits represent a NIC (Network Interface Controller).														

Example MAC Address

3A-34-52-C4-69-B8

Organizationally
Unique Identifier
(OUI)

Network Interface
Controller
(NIC)

MAC addresses play a crucial role in packet routing at the data link layer of the OSI model. When a packet is sent from one device to another over a local network, it is encapsulated with a header that contains the MAC address of the source and destination devices.

When the packet reaches the local network, the network switches use the destination MAC address to determine which device to forward the packet to. The switch checks its MAC address table to see if it knows the location of the device with that MAC address, and if it does, it sends the packet directly to that device. If it doesn't, the switch will broadcast the packet to all devices on the network in order to find the destination device.

Once the destination device receives the packet, it uses the MAC address in the header to identify the packet as being intended for itself and then extracts the data from the packet. Therefore, MAC addresses are essential in local network communication, allowing for the efficient routing of packets to their intended destination.

12.6.2.3 identify the MAC address of a computer

To identify the MAC address of a computer, you can follow these steps:

1. On Windows, click on the Start menu and type "**cmd**" into the search box to open the Command Prompt. On a Mac, open the Terminal app.
2. In the Command Prompt or Terminal, type the command "**ipconfig /all**" (on Windows) or "**ifconfig**" (on Mac).
3. Look for the network adapter you want to find the MAC address for. The MAC address will be listed under the "**Physical Address**" or "**HWaddr**" section.
4. The MAC address is a series of six pairs of alphanumeric characters, separated by colons or dashes. For example, "00-1a-2b-3c-4d-5e".